Gas, wind and *the industrial energy fight*

Wind is winning power generation. The harder contest, and the bigger prize, is the energy European industry uses: volumes, costs, the 2022 price shock, and why industry is harder to decarbonise than power.

31%

of EU industry's final energy is natural gas (2022)

€340/MWh

the gas price at its crisis peak, August 2022

€180–220

per tonne of CO_2 avoided: the extra cost of green steel and green ammonia today

THE VOLUME PICTURE

What does gas actually do in EU industry?

≈ 2,600–2,800 TWh

EU industry's final energy use a year (2022–2023). Eurostat reports it in petajoules: 1 TWh = 3.6 PJ.

31.2% gas

about 870 TWh in 2022, the second-largest source

33.3% electricity

the largest source, mostly motors, pumps and low-temperature processes

The rest

oil, coal, biomass and heat

Eurostat energy balances, final energy consumption in industry. The original Moodle slide said ~400 TWh with 40% gas, which is far too small: replacing industrial gas is a much larger task than that implied.

THE PRICE STORY, 2019–2024

Gas against wind

	€ PER MWH
Gas, 2019: the pre-crisis norm	≈ 18
Gas, August 2022: the crisis peak	≈ 340
Gas, 2024: normalised	≈ 35
Wind power purchase agreement, 10 years, 2024	55–75

The crisis lesson

At €340, energy became an existential cost: German chemical plants cut output, fertiliser producers shut down.

Why wind looked cheap

Fixed-price wind contracts at €55–75 drove a wave of industrial PPAs in 2022–23.

The normalisation problem

With gas back near €35, industrial electricity at about €85 is again dearer for direct heat. The crisis made the case; normalisation softened it.

THE HARD-TO-ABATE PROBLEM

Where gas cannot simply be replaced

The difficulty is thermodynamic and economic: high-temperature heat and chemical feedstocks cannot be replaced by electrons alone.

How hard, by sector

Hard (feedstock or heat above 400 °C): chemicals, iron and steel, cement, glass.
Moderate: paper and pulp, food and beverage.
Electrifiable now: aluminium, machinery.

The green premium · € per tonne of CO_2 avoided

Green ammonia (electrolysis) **220** · green steel (hydrogen) **180** · industrial heat above 400 °C (hydrogen boiler) **140** · cement with carbon capture **95** · food and beverage steam **25** · heat below 200 °C (heat pump) **15** · aluminium **5**

The EU carbon price averaged about €65 a tonne in 2024: enough for low-temperature heat, far from enough for steel or ammonia.

IEA Tracking Industry 2023; BloombergNEF Green Steel Tracker; Agora Energiewende, industrial heat, 2023, as compiled in the original Moodle deck.

STRATEGIC IMPLICATIONS

What this means for Ørsted and Eni

Ørsted · the industrial opportunity

Industrial PPAs with energy-intensive manufacturers: a revenue pillar and a hedge against market prices.
Green hydrogen (offshore wind + electrolysis) is the long game, if its cost falls on a learning curve.
The timing risk: industry needs supply certainty before it retrofits plants; project delays cost credibility.
The scale: replacing EU industrial gas with green hydrogen would need ~800–1,000 GW of offshore wind, **24–30 times** EU offshore capacity today (~33 GW).

Eni · the last stronghold and its limits

Industrial gas is gas's best remaining argument: feedstock and high-temperature heat need it longest, and Eni's gas and LNG business serves it.
Carbon pricing closes the window as the ETS rises toward €100–130/t.
Carbon capture is Eni's bridge: decarbonise industry without abandoning gas.
The capital test: every euro in new gas infrastructure is a euro not in Plenitude or hydrogen.

Consulting hypothesis: the industrial transition will be decided by green hydrogen costs and the carbon price, not by the offshore wind learning curve alone.

WRIGHT'S LAW

Why the cost keeps falling

The rule

Every time cumulative production doubles, cost falls by a fixed percentage, the learning rate. It does not matter whether the doubling takes two years or ten.

Why it compounds

Cheaper technology → more deployment → another doubling → another cost cut. A reinforcing loop.

Deployment is the R&D

Technologies deployed at scale get much cheaper; those that are not stagnate. Policy that drives deployment drives cost down.

Offshore wind: about 11–15% per doubling. Solar PV: about 23%. New nuclear: about zero, or negative.

You fit the offshore rate yourself in Week 3.

LCOE BY TECHNOLOGY · GLOBAL, 2024

What a MWh costs to produce

TECHNOLOGY	TYPICAL, $/MWH	RANGE UP TO
Solar PV, utility scale	**48**	**120**
Onshore wind	**50**	**102**
Large hydro	**60**	**200**
Offshore wind	**84**	**160**
Gas, combined cycle	**90**	**145**
Coal, new	**105**	**185**
Gas peaker · new nuclear	**160 · 160**	**250 · 310**

Levelised cost: the lifetime average cost per MWh, construction, financing, fuel and operation combined. IRENA 2023; IEA WEO 2024, as compiled in the original deck. Onshore wind and solar beat new gas almost everywhere; offshore wind is now about level with new gas on the typical value, and gas carries a fuel-price risk that LCOE does not show.

OFFSHORE WIND · 2010 TO 2024

The cost trajectory, and the 2022–23 shock

	2010	2021 · THE LOW	2023 · THE SHOCK	2024
Offshore wind LCOE, $/MWh	188	75	96	86
Fall since 2010	—	60%	49%	54%

The long run

Larger turbines, a learning supply chain and experienced developers: a fall of 54–60%, not the "~70%" the original deck claimed.

The shock

Inflation, interest rates and supply chains. Ørsted booked DKK 28.4 bn of impairments in the first nine months of 2023 on US projects priced before inflation.

The recovery

The curve resumed in 2024: the interruption looks cyclical. Week 4 shows why: most of the cost of offshore wind is the cost of money.

THE STRATEGIC TENSION

What falling LCOE means for Ørsted and Eni

Ørsted · riding the curve

Wright's Law is the business model: every cost fall validates the pivot from fossil fuels.

A reinforcing loop: lower LCOE → more viable projects → more scale → lower LCOE.

2022–23 exposed the execution risk: projects at fixed pre-inflation prices became loss-making; the cost curve is not project profit.

The open question: can it keep going in deeper water and harder markets?

Eni · straddling the divide

Falling LCOE erodes the fossil case: industrial buyers prefer fixed-price renewable contracts to volatile gas.

Stranded-asset risk: gas infrastructure is a 20–30-year investment and may be uncompetitive before it is paid off.

The hedge: gas as a "transition fuel" holds today; to 2035 depends on the curve.

The core question: can Plenitude grow as fast as costs fall, or do fossil returns crowd it out?

Consulting hypothesis: Ørsted's risk is execution at scale in adverse markets; Eni's is a strategic identity that is hard to sustain if the cost curve continues for another decade.